DATA SCIENTIST

Learning Timetable

8-Week Intensive Program

Program Overview

Duration: 8 Weeks (2 Months)

Daily Hours: 9:00 AM - 5:00 PM (8 Hours)

Days: Monday - Saturday

Sunday: Review & Practice

Skills You Will Learn:

- Python Programming (Beginner to Intermediate)
- SQL & Database Management
- Pandas & NumPy for Data Manipulation
- Data Cleaning & Exploratory Data Analysis
- Machine Learning Fundamentals

- Model Evaluation & Feature Engineering
- End-to-End ML Project Development

8-Week Program Overview

Phase	Weeks	Focus Area	Key Topics
Phase 1	Weeks 1-2	Python Fundamentals	Variables, data types, control flow, functions, OOP basics, file I/O, error handli
Phase 2	Weeks 3-4	SQL & Databases	SELECT, WHERE, JOINs, GROUP BY, subqueries, window functions, Python
Phase 3	Weeks 5-6	Pandas & NumPy	Arrays, dataframes, indexing, filtering, merging, grouping, missing data, EDA
Phase 4	Weeks 7-8	Machine Learning	Regression, classification, clustering, model evaluation, feature engineering, fi

MILESTONE: Complete all 4 phases to become job-ready for junior data scientist roles

Phase 1: Python Fundamentals (Weeks 1-2)

Week 1: Python Basics

Monday

9:00 AM - 10:30 AM Variables, Data Types (int, float, str, bool)

10:30

Tuesday

9:00 AM - 10:30 AM Dictionaries - creation, access, methods

10:30

Wednesday

9:00 AM - 10:30 AM Loops - for loops, range(), enumerate()

10:30

Thursday

9:00 AM - 10:30 AM Functions - definition, parameters, return values

10:30

Friday

9:00 AM - 10:30 AM

	File I/O - reading/writing text files

10:30

Saturday

9:00 AM - 10:30 AM	Review: All Week 1 concepts

10:30

Sunday

9:00 AM - 12:00 PM	Review weak areas, redo difficult problems

12:00

Week 2: Intermediate Python

Monday

9:00 AM - 10:30 AM OOP - Classes and Objects

10:30 A

Tuesday

9:00 AM - 10:30 AM Regular Expressions (re module)

10:30 A

Wednesday

9:00 AM - 10:30 AM Working with APIs (requests library)

10:30 A

Thursday

9:00 AM - 10:30 AM Virtual environments, Jupyter Notebook

10:30 A

Friday

9:00 AM - 10:30 AM Matplotlib & Seaborn basics

10:30 A

Saturday

9:00 AM - 10:30 AM	Review: All Week 2 concepts

10:30

Sunday

9:00 AM - 12:00 PM	Review weak areas, redo difficult problems

12:00

MILESTONE: Comfortable writing 100+ line Python programs with OOP

Phase 2: SQL & Databases (Weeks 3-4)

Week 3: SQL Fundamentals

Monday

9:00 AM - 10:30 AM Introduction to Databases, Relational vs NoSQL

10:30 A

Tuesday

9:00 AM - 10:30 AM ORDER BY, LIMIT, DISTINCT

10:30 A

Wednesday

9:00 AM - 10:30 AM Aggregate functions: COUNT, SUM, AVG, MIN, MAX

10:30 A

Thursday

9:00 AM - 10:30 AM JOINs - INNER JOIN

10:30 A

Friday

9:00 AM - 10:30 AM

	Subqueries - WHERE clause subqueries

10:30

Saturday

9:00 AM - 10:30 AM	Window functions: ROW_NUMBER, RANK, DENSE_RANK

10:30

Sunday

9:00 AM - 12:00 PM	Review all SQL concepts, redo weak areas

12:00

Week 4: Advanced SQL + Python Integration

Monday

9:00 AM - 10:30 AM Indexing and query optimization

10:30 A

Tuesday

9:00 AM - 10:30 AM Python + SQL: sqlite3 library

10:30 A

Wednesday

9:00 AM - 10:30 AM SQLAlchemy ORM basics

10:30 A

Thursday

9:00 AM - 10:30 AM Advanced: Transactions, constraints

10:30 A

Friday

9:00 AM - 10:30 AM Review: All SQL concepts

10:30 A

Saturday

9:00 AM - 10:30 AM	Python + SQL integration review

10:30

Sunday

9:00 AM - 12:00 PM	Review weak areas, redo difficult problems

12:00

MILESTONE: Can write complex SQL queries and connect Python to databases

Phase 3: Pandas & NumPy (Weeks 5-6)

Week 5: NumPy

Monday

9:00 AM - 10:30 AM NumPy arrays vs Python lists

10:30 A

Tuesday

9:00 AM - 10:30 AM Array operations: +, -, *, /, dot product

10:30 A

Wednesday

9:00 AM - 10:30 AM Array reshaping: reshape(), ravel(), flatten()

10:30 A

Thursday

9:00 AM - 10:30 AM Random number generation

10:30 A

Friday

9:00 AM - 10:30 AM

	Linear algebra: matrix operations

10:30

Saturday

9:00 AM - 10:30 AM	Review: All NumPy concepts

10:30

Sunday

9:00 AM - 12:00 PM	Review weak areas, redo difficult problems

12:00

Week 6: Pandas

Monday

9:00 AM - 10:30 AM Series and DataFrames, head(), info(), describe()

10:30

Tuesday

9:00 AM - 10:30 AM Filtering with boolean indexing

10:30

Wednesday

9:00 AM - 10:30 AM groupby() and aggregation

10:30

Thursday

9:00 AM - 10:30 AM Working with dates and times

10:30

Friday

9:00 AM - 10:30 AM Reading/writing CSV, Excel, JSON

10:30

Saturday

9:00 AM - 10:30 AM Review: All Pandas concepts

10:30

Sunday

9:00 AM - 12:00 PM Review weak areas, redo difficult problems

12:00

MILESTONE: Proficient in data wrangling with Pandas and NumPy

Phase 4: Machine Learning (Weeks 7-8)

Week 7: ML Foundations

Monday

9:00 AM - 10:30 AM What is ML? Supervised vs Unsupervised learning

10:30

Tuesday

9:00 AM - 10:30 AM Linear Regression - theory

10:30

Wednesday

9:00 AM - 10:30 AM Logistic Regression - theory

10:30

Thursday

9:00 AM - 10:30 AM Evaluation metrics: accuracy, precision, recall, F1

10:30

Friday

9:00 AM - 10:30 AM

	K-Nearest Neighbors (KNN)

10:30

Saturday

9:00 AM - 10:30 AM	Random Forests - theory

10:30

Sunday

9:00 AM - 12:00 PM	Review all ML concepts, redo weak areas

12:00

Week 8: Advanced ML + Final Project

Monday

9:00 AM - 10:30 AM Hyperparameter tuning: GridSearchCV, RandomizedSearchCV

10:30

Tuesday

9:00 AM - 10:30 AM Feature scaling: StandardScaler, MinMaxScaler

10:30

Wednesday

9:00 AM - 10:30 AM K-Means implementation, elbow method

10:30

Thursday

9:00 AM - 10:30 AM Final Project: Choose a dataset (Kaggle)

10:30

Friday

9:00 AM - 10:30 AM Final Project: Model evaluation and tuning

10:30

Saturday

9:00 AM - 10:30 AM	Final Project: Complete and polish

10:30

Sunday

9:00 AM - 12:00 PM	Celebrate! Review your progress

12:00

MILESTONE: Complete an end-to-end ML portfolio project from scratch

Recommended Free Resources

Topic	Resources
Python	Automate the Boring Stuff (automatetheboringstuff.com) | Python.org Tutorial | Corey Schafer YouTube
SQL	SQLBolt (sqlbolt.com) | Mode SQL Tutorial | HackerRank SQL Practice
Pandas	Kaggle Pandas Course | 10 Minutes to Pandas | Pandas documentation
NumPy	NumPy Quickstart | Kaggle NumPy | NumPy documentation
Machine Learning	Google ML Crash Course | Andrew Ng ML Specialization (Coursera) | Kaggle Learn
Practice	LeetCode (Python) | HackerRank | Kaggle Competitions
Datasets	Kaggle Datasets | UCI ML Repository | Google Dataset Search

Tips for Success

1. Code every day - consistency beats cramming
2. Type code manually - do not just copy/paste
3. Build projects - they matter more than certificates
4. Use flashcards (Anki) for SQL syntax and Pandas methods
5. Join communities - r/datascience, Kaggle forums, Discord servers
6. Teach what you learn - write blog posts or explain to a friend
7. Take breaks - avoid burnout with the Pomodoro technique
8. Track progress - keep a learning journal
9. Ask questions - use Stack Overflow and Reddit
10. Stay curious - explore topics beyond the curriculum

Daily Schedule Template (9 AM - 5 PM)

Time	Activity
9:00 AM - 10:30 AM	Theory / Concept Learning
10:30 AM - 10:45 AM	Short Break
10:45 AM - 12:30 PM	Hands-on Practice / Coding
12:30 PM - 1:30 PM	Lunch Break
1:30 PM - 3:00 PM	Guided Practice / Exercises

3:00 PM - 3:15 PM	Short Break
3:15 PM - 5:00 PM	Projects / Review / Practice